

Finally, we can generalize the first trick if the right-hand side is an exponential, e^{at} , times a polynomial of degree n , $P_n(t)$. Then the particular solution of

$$(D - b_1)(D - b_2)x = e^{at}P_n(t)$$

is

$$x_p = \begin{cases} e^{at}Q_n(t) & \text{if } a \neq b_1 \text{ and } a \neq b_2 \\ te^{at}Q_n(t) & \text{if } a = b_1 \text{ or } a = b_2 \text{ and } b_1 \neq b_2 \\ t^2e^{at}Q_n(t) & \text{if } a = b_1 = b_2 \end{cases},$$

where $Q_n(t)$ is a polynomial of degree n :

$$Q_n(t) = A + Bt + Ct^2 + \dots$$

To determine $A, B, C, \dots$ you plug x_p into the differential equation and equate the coefficients of the different powers of t .

Worked Example 11.4 Find the general solution for

$$(D^2 - 4)x = 2 - 8t$$

when $t = 0$, $x = 0$, and $\dot{x} = 5$.

Solution We use the method described by [Equation \(11.21\)](#) and the discussion following it. We write the differential equation as

$$(D - 2)(D + 2)x = 2 - 8t.$$

Let $u = (D + 2)x$ and write

$$(D - 2)u = 2 - 8t.$$

For a first-order differential equation such as

$$(D + f(t))x = g(t),$$

the integrating factor is $e^{\int f(t)dt} = e^{-2t}$. Multiplying by e^{-2t} we obtain

$$e^{-2t} \frac{du}{dt} - 2ue^{-2t} = 2e^{-2t} - 8te^{-2t},$$

$$\frac{d}{dt}(ue^{-2t}) = 2e^{-2t} - 8te^{-2t},$$

$$\int d(ue^{-2t}) = \int 2e^{-2t} dt - \int 8te^{-2t} dt.$$

Therefore,

$$u = C_1 e^{2t} + 4t + 1.$$

But since $u = (D + 2)x$ we have

$$\frac{dx}{dt} + 2x = C_1 e^{2t} + 4t + 1.$$

Now the integrating factor is e^{+2t} so we multiply through by this factor to get

$$\frac{d}{dt}(xe^{2t}) = C_1 e^{4t} + 4te^{2t} + e^{2t},$$

and consequently,

$$x(t) = C_2 e^{-2t} + \frac{1}{4} C_1 e^{2t} + 2t - \frac{1}{2}.$$

Inserting the initial conditions leads to $C_1 = 4$ and $C_2 = -1/2$, and finally

$$x(t) = e^{2t} - \frac{1}{2} e^{-2t} + 2t - \frac{1}{2}.$$

Note that this method yields the complete general solution, that is, it gives us the sum of the particular solution and the complementary function all at the same time.

Exercise 11.19

Obtain the general solution to $(D^2 - D - 6)x = 8$, where $D \equiv \frac{d}{dt}$. Answer:

$$x(t) = C_1 e^{3t} + C_2 e^{-2t} - \frac{4}{3}.$$

Exercise 11.20

Obtain the general solution to $(D^2 - 9)x = 5e^{-2t}$ assuming that $x(t = 0) = 0$ and $\dot{x}(t = 0) = 1$ (in the appropriate set of units). Answer:
 $x(t) = \frac{1}{3}e^{3t} + \frac{2}{3}e^{-3t} - e^{-2t}$.

Exercise 11.21

Obtain the general solution of $(D^2 - 4)x = \sin t$. Answer:
 $x(t) = C_1e^{2t} + C_2e^{-2t} - \frac{1}{5}\sin t$.

11.4.4 The Forced Undamped Oscillator

Now that you have seen some methods for solving inhomogeneous second-order differential equations, let us return to the problem of the forced undamped harmonic oscillator.

The differential equation we set out to solve was [Equation \(11.19\)](#):

$$\ddot{x} + \omega_0^2 x = \frac{F_0}{m} \sin \omega_d t.$$

The right-hand side of this equation is proportional to $\sin \omega_d t$ so we use trick #2 and write

$$(D - i\omega_0)(D + i\omega_0)\xi_p(t) = \frac{F_0}{m} e^{i\omega_d t}. \quad (11.25)$$

According to trick #2 and [Equation \(11.23\)](#) the particular solution of [Equation \(11.25\)](#) is

$$\xi_p(t) = Ce^{i\omega_d t}.$$

To obtain C , plug ξ_p back into the differential equation, thus:

$$\left(\frac{d^2}{dt^2} + \omega_0^2 \right) Ce^{i\omega_d t} = \frac{F_0}{m} e^{i\omega_d t}.$$

Hence,

$$-\omega_d^2 C + \omega_0^2 C = \frac{F_0}{m},$$

and

$$C = \frac{F_0/m}{\omega_0^2 - \omega_d^2}$$

Recall that x_p is the imaginary part of the complex function ξ_p , so the particular solution is

$$x_p(t) = \mathcal{I}m(\xi_p(t)) = \mathcal{I}m\left(\frac{F_0/m}{\omega_0^2 - \omega_d^2} e^{i\omega_d t}\right) = \frac{F_0/m}{\omega_0^2 - \omega_d^2} \sin \omega_d t,$$

and the general solution of Equation (11.19) is

$$x(t) = A \cos(\omega_0 t + \theta_0) + \frac{F_0/m}{\omega_0^2 - \omega_d^2} \sin \omega_d t.$$

The problem is now solved, except for determining A and θ_0 from the initial conditions.

Before leaving this problem, I want you to notice that there is something quite interesting about the particular solution,

$$x_p = \frac{F_0/m}{\omega_0^2 - \omega_d^2} \sin \omega_d t.$$

This expression indicates that as the driving frequency ω_d gets closer and closer to ω_0 , the natural frequency of oscillation, the amplitude of the oscillations get larger and larger. If $\omega_d = \omega_0$, the amplitude is infinite. This phenomenon is known as *resonance*. We will consider resonance further in the next section, refining our analysis and showing that the amplitude does not actually become infinite for a real physical system.

Exercise 11.22

A simple harmonic oscillator consists of a mass of 3 kg on a spring of force constant 0.15 N/m. Determine the frequency of the driving force which will cause the amplitude of oscillations to grow extremely large. Answer 0.22 rad/s.

11.4.5 The Forced Damped Oscillator

We now consider a *damped*, driven, harmonic oscillator and assume the driving force is, once again, $F_0 \sin \omega_d t$. The equation of motion is

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = \frac{F_0}{m}\sin \omega_d t.$$

Note that $\sqrt{k/m} = \omega_0$ is the "natural" frequency of the undamped oscillator. For the sake of a specific example, we assume the oscillator is underdamped.

The technique for finding the particular solution is the same as before, so set

$$\xi_p = Ce^{i\omega_d t},$$

and plug it into the differential equation. This gives

$$C(-\omega_d^2)e^{i\omega_d t} + \frac{b}{m}C(i\omega_d)e^{i\omega_d t} + \omega_0^2 Ce^{i\omega_d t} = \frac{F_0}{m}e^{i\omega_d t},$$

where $\sin \omega_d t$ was replaced by $e^{i\omega_d t}$. (This means that at the end of the procedure we must only keep the imaginary part of the solution.)

Solving for C yields

$$C = \frac{F_0/m}{(\omega_0^2 - \omega_d^2) + i\omega_d b/m},$$

which is complex. To make this quantity easier to manipulate, the imaginary term is usually placed in the numerator. To achieve this, multiply top and bottom by the complex conjugate of the denominator and obtain

$$C = \frac{(F_0/m)[(\omega_0^2 - \omega_d^2) - i\omega_d b/m]}{[(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2]}.$$

You have already seen that the general solution of the homogeneous equation for the underdamped harmonic oscillator is given in [Equation \(11.15\)](#) as $x = Ae^{-\gamma t} \cos(\omega_1 t + \theta)$. This is the complementary function for the

inhomogeneous equation. The general solution of the forced, underdamped harmonic oscillator is, then,

$$x(t) = Ae^{-\gamma t} \cos(\omega_1 t + \theta) + \mathcal{I}m \left(\frac{F_0/m[(\omega_0^2 - \omega_d^2) - i\omega_d b/m]}{[(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2]} e^{i\omega_d t} \right).$$

Eventually, the first term will die out. (This is, of course, also true for overdamped and critically damped oscillators.) Dropping the first term, you are left with

$$x(t) = \mathcal{I}m \left(\frac{F_0/m[(\omega_0^2 - \omega_d^2) - i\omega_d b/m]}{[(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2]} (\cos \omega_d t + i \sin \omega_d t) \right).$$

After a bit of algebra, this expression can be written

$$x(t) = \frac{F_0}{m} \frac{(\omega_0^2 - \omega_d^2) \sin \omega_d t - (\omega_d b/m) \cos \omega_d t}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}. \quad (11.26)$$

At resonance, when $\omega_0 = \omega_d$, the denominator is minimized, but the damping term prevents it from going to zero. Therefore, at resonance you get a large amplitude, but it is not infinite. In real physical systems, there is always some damping that prevents the system from having infinite amplitude at resonance.

Equation (11.26) can be expressed in a more convenient way by using the trigonometric identity

$$A \sin \omega t + B \cos \omega t = \sqrt{A^2 + B^2} \cos(\omega t - \phi),$$

where $\phi = \tan^{-1}(A/B)$. So,

$$\begin{aligned} x(t) &= \left[\frac{(F_0/m)^2 [(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2/m^2]}{[(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2/m^2]^2} \right]^{1/2} \cos(\omega_d t - \phi) \\ &= \frac{F_0}{[m^2(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2]^{1/2}} \cos(\omega_d t - \phi), \end{aligned} \quad (11.27)$$

where, from Equation (11.26),

$$\phi = \tan^{-1} \left[-\frac{\omega_0^2 - \omega_d^2}{\omega_d b/m} \right].$$

It is interesting to note that the amplitude of $x(t)$ is not maximized at $\omega_d = \omega_0$ but rather at a nearby frequency called the resonant frequency which you can determine by evaluating the derivative of the denominator and setting it equal to zero, thus:

$$\frac{d}{d\omega_d} [m^2(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2]^{1/2} = 0.$$

This leads to the following condition for a minimum

$$\omega_d^2 = \omega_0^2 - \frac{b^2}{2m^2}. \quad (11.28)$$

The greatest amplitude occurs when the driving frequency is equal to the resonant frequency which we can denote by ω' .

$$\omega' = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}.$$

This is neither the frequency of the undamped oscillator (ω_0), nor the frequency of the damped oscillator (ω_1).

When $\omega_d = \omega'$ the amplitude of the oscillation is given by

$$A = \frac{F_0}{[m^2(\omega_0^2 - \omega'^2)^2 + \omega'^2 b^2]^{1/2}}.$$

The resonance phenomenon is often associated with the transfer of energy. The average energy of a harmonic oscillator is proportional to the amplitude squared, so an important and useful parameter in studying resonance is A^2 , given by

$$A^2 = \frac{F_0^2}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 b^2}.$$

Worked Example 11.5 Show that the phase angle ϕ in Equation (11.27) is given by $\phi = \tan^{-1} \left[\frac{\omega_0^2 - \omega_d^2}{\omega_d b/m} \right]$.

Solution We have shown in Equation (11.26) that

$$x = x(t) = \frac{F_0}{m} \frac{(\omega_0^2 - \omega_d^2) \sin \omega_d t - (\omega_d b/m) \cos \omega_d t}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}.$$

Writing this in the form $A \sin \omega t + B \cos \omega t$ we have

$$A = \frac{F_0}{m} \frac{(\omega_0^2 - \omega_d^2)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2},$$

and

$$B = \frac{F_0}{m} \frac{-(\omega_d b/m)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}.$$

So $\phi = \tan^{-1}(\mathcal{A}/\mathcal{B})$ yields

$$\begin{aligned}\phi &= \tan^{-1} \frac{(\omega_0^2 - \omega_d^2)}{-(\omega_d b/m)} = \tan^{-1} \frac{(\omega_0^2 - \omega_d^2)}{-(\omega_d b/m)} \\ &= \tan^{-1} \frac{-(\omega_d^2 - \omega_0^2)}{\omega_d b/m}\end{aligned}$$

The phase shift ϕ describes the lag between the applied force $F(t)$ and the response $x(t)$.

Exercise 11.23

Fill in the missing steps in the derivation of [Equation \(11.28\)](#).

Exercise 11.23

Show that $A \sin \omega t + B \cos \omega t = \sqrt{A^2 + B^2} \cos(\omega t - \phi)$, where $\tan \phi = A/B$.

Exercise 11.25

A damped harmonic oscillator is driven by a force $F_0 e^{-\alpha t}$. Find the general solution. (Note: The particular solution can be expected to have the same time dependence as the driving force.)

The Q Factor

The Q factor is an important property of a damped driven harmonic oscillator. It is related to both the shape of the resonance curve and the energy loss rate of the oscillator.⁶

There are several related definitions of the Q factor in terms of the system. These yield similar but not identical results. You should be aware that in specific applications the Q factor may be expressed differently than the way we will describe here, but the basic ideas are always the same.

Let us begin with the definition of the Q factor in terms of the *shape* of the resonance curve. An appropriate definition of Q is the ratio of the resonant frequency (ω_0) to the width of the resonance curve at half maximum of the curve. ($\Delta\omega$ is denoted "full width at half maximum.") That is, $Q = \frac{\omega_0}{\Delta\omega}$ which, as we shall show, can be written as

$$Q \equiv \frac{m}{b} \omega_0 = \sqrt{\frac{km^2}{b^2}}.$$

Note that the definition only involves the physical properties of the oscillator. As mentioned previously, a driven oscillator will have the largest amplitude when the frequency of the driving force is equal to the resonant frequency, which we have denoted ω' . For simplicity, we shall consider systems with small damping values, so $\omega' \doteq \omega_0 = \sqrt{k/m}$. A plot of the square

⁶ There are many applications of driven harmonic oscillators. For example, in musical instruments such as trumpets or French horns the vibration of a reed or of the musician's lips produces sounds with a wide range of frequencies. The instrument has a resonance at the desired frequency, thus picking out the desired note. In optical devices, such as the resonant cavity of lasers, the Q factor is an important parameter. Electromechanical resonators such as quartz crystals in digital watches allow for highly accurate timepieces.

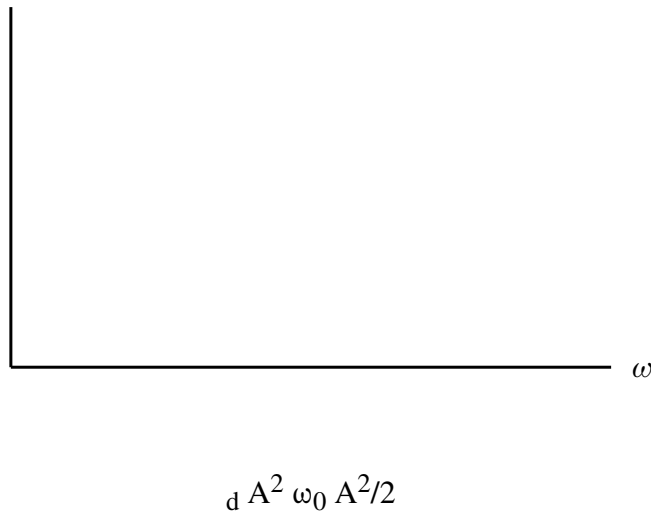


Figure 11.8 The resonance curve with full width at half maximum indicated.

of the amplitude as a function of driving frequency (ω_d) will look somewhat as illustrated in [Figure 11.8](#).

To obtain a large response at a specific frequency the curve of [Figure 11.8](#) should be very tall and very thin. In technical terms, it is desirable for the curve to have a small value for the width at half maximum (as shown by the double-headed arrow in the figure).

We have presented the resonance curve in terms of the amplitude squared rather than the amplitude. This is convenient because the energy of the oscillator is proportional to the square of the amplitude. Furthermore, a slightly different definition of Q is the ratio of the maximum energy stored in the oscillator to the energy lost per radian of motion. (We explore this alternative definition in [Problem 11.16](#).)

The square of the amplitude is a maximum at $\omega_d = \omega_0$ and is given by

$$A_{\omega_0}^2 = \frac{F_0^2}{\omega_0^2 b^2}.$$

For some nearby driving frequency ω the amplitude is

$$A_{\omega}^2 = \frac{F_0^2/m^2}{[(\omega_0^2 - \omega^2)^2 + \omega^2 b^2/m^2]}.$$

If $A_{\omega}^2 = \frac{1}{2} A_{\omega_0}^2$ we have

$$2\omega_0^2 b^2 / m^2 = (\omega_0^2 - \omega^2)^2 + \omega^2 b^2 / m^2,$$

$$\frac{b^2}{m^2}(2\omega_0^2 - \omega^2) = (\omega_0^2 - \omega^2)^2,$$

and for ω close to ω_0 , we can write the approximate relation

$$\frac{b}{m}\omega_0 = \omega_0^2 - \omega^2.$$

Now $(\omega_0^2 - \omega^2) = (\omega_0 - \omega)(\omega_0 + \omega) \approx \Delta\omega_{1/2}(2\omega_0)$, where $\Delta\omega_{1/2} = \omega - \omega_0$ is the half width at half maximum. The full width at half maximum is $\Delta\omega = 2\Delta\omega_{1/2}$, so

$$\frac{b\omega_0}{m} = 2\Delta\omega_{1/2} \cdot \omega_0 = \Delta\omega\omega_0,$$

and

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0}{b/m} = \frac{m}{b}\omega_0.$$

You can appreciate that large Q means a small value of $\Delta\omega$, that is, a thin peak. It might be mentioned that a very large Q is not always desirable, as for example in a musical instrument it would make it harder to "hit the note." On the other hand, for a radio to tune into one of two stations broadcasting at nearby frequencies, a high Q factor is highly desirable.

Another use of the Q factor is in determining the loss of energy of a damped oscillator. For simplicity let us consider an unforced oscillator. As you know, due to damping, the amplitude of the oscillations will decay exponentially as $e^{-\gamma t}$, where $\gamma = b/2m$; see [Equation \(11.15\)](#). Energy is proportional to the amplitude squared, so the decrease in energy in time t is given by $(e^{-\gamma t})^2 = e^{-2\gamma t}$. Therefore, the energy loss in the time the oscillator goes through one complete cycle is $e^{-2\gamma T}$, where T is the period ($T = \frac{2\pi}{\omega_0}$). Consequently, the ratio of final energy to initial energy is

$$\frac{Ee^{-2\gamma T}}{E} = e^{-2\gamma T} = e^{-2\frac{b}{2m}\frac{2\pi}{\omega_0}} = e^{-2\pi\frac{b}{m\omega_0}} = e^{-2\pi/Q}.$$

This tells us that an oscillator with large Q does not lose energy as quickly as an oscillator with small Q .

Resonance in Electrical Circuits (Optional)

Resonance is an interesting physical phenomenon of importance in such diverse fields as the design of electrical circuits and the motion of celestial bodies. For example, the RLC circuit you studied in your introductory electricity course is illustrated in [Figure 11.9](#).

The charge on the capacitor plates is q . If the plates are initially charged, there will be a flow of charge (a current, I) through the inductor, setting up a back emf or voltage difference given by $L \frac{dI}{dt}$. This can also be written as $L \frac{d^2q}{dt^2}$. The voltage drop across the resistor is IR and this can be written as $R \frac{dq}{dt}$. Finally, the voltage drop across the capacitor is given by $\frac{q}{C}$. The sum of voltage drops around the circuit is zero, so

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C}q = 0.$$

But this is exactly the equation for a damped harmonic oscillator! If you add a "driving force" by placing some source of emf in the circuit (such as an oscillator or power supply) you will have a driven oscillator. The natural frequency of the circuit is $\omega_0 = \sqrt{1/LC}$, as you can appreciate from the form of the equation if the damping term is removed. If the driving frequency is near or equal to the natural frequency, the amplitude of the current increases greatly. This is the principle behind tuning a radio or television receiver. The "driving" voltage is supplied by the antenna picking up electromagnetic radiation. Suppose a station you want to tune to is broadcasting at a frequency ω_d . The receiver contains a variable capacitor whose capacitance you change by turning a knob or pushing a button on your receiver. Changing C will, of course, change the value of ω_0 and hence the resonant frequency ω' . When ω' is equal to ω_d , the current through the resistor reaches a much higher value than for other frequencies. This is illustrated in [Figure 11.10](#).

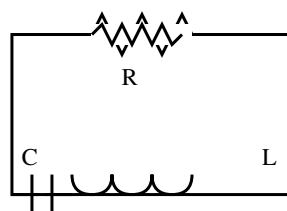
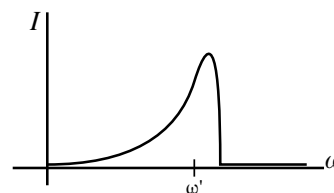


Figure 11.9 An RLC circuit.

Figure 11.10 A plot of the current (I) as a function of the angular frequency ω , illustrating the resonance phenomenon in an LRC circuit. When the frequency ω is nearly equal to the resonant frequency ω' , the current in the circuit increases greatly. If there were no resistance in the circuit, the current would become infinite at $\omega = \omega_0$.



Resonance in Astronomy

As mentioned in [Chapter 10](#), celestial bodies are affected by resonances. For example, the orbital period and the rotational period of the Moon are "locked" into a 1:1 resonance. In celestial mechanics, a resonance is defined as a system in which the frequency of the "driving" force (usually the effect of a perturbing body) and the "natural" frequency of the system are in the ratio of small numbers. Resonances explain why there are gaps in the rings of Saturn and in the asteroid belts between Mars and Jupiter, and why certain satellites (including the Moon) have rotation rates equal to their orbital rates. (The resonance phenomenon in celestial mechanics is often called a "commensurability.")

Exercise 11.26

An RLC series circuit consists of a 10 ohm resistor, a 6 microfarad capacitor, and an 0.2 henry inductor. Determine the resonant frequency for the circuit.

Answer: 913 rad/s.

11.5 Coupled Oscillators

In the real physical world, oscillators are seldom isolated from their environment and an oscillating body will set up oscillations in nearby bodies. A very good example of *coupled oscillators* is the motion of atoms in a crystal. A vibrating atom will cause neighboring atoms to start oscillating.

The analysis of coupled oscillators can be quite complicated. This section is a study

$$\ddot{x}_1 + (k'/m)x_1 - (k_3/m)x_2 = 0,$$

$$\ddot{x}_2 + (k'/m)x_2 - (k_3/m)x_1 = 0.$$

Now replace k'/m by ω_0^2 . The reason for this notation is that ω_0 is the frequency of oscillation of either one of the masses if the other mass is held at rest. Furthermore, let us denote k_3/m by $\Delta\omega^2$, for reasons that will soon become clear.

Using this new notation the equations become

$$\ddot{x}_1 + \omega_0^2 x_1 - \Delta\omega^2 x_2 = 0, \quad (11.30)$$

$$\ddot{x}_2 + \omega_0^2 x_2 - \Delta\omega^2 x_1 = 0.$$

This system of *coupled* linear differential equations can be solved by assuming the solutions

$$x_1 = Ae^{i\omega t},$$

$$x_2 = Be^{i\omega t}.$$

We assume oscillatory solutions (that is why the exponent is imaginary) with both of the variables x_1 and x_2 having the same angular frequency ω . (Except for a slight change in the notation, this is the same technique previously used in assuming solutions of the form e^{pt} .) Following our usual procedure we insert the expressions for x_1 and x_2 into the coupled equations of motion to determine the "auxiliary equations" that will give us ω . We obtain

$$A(i\omega)^2 e^{i\omega t} + \omega_0^2 A e^{i\omega t} - \Delta\omega^2 B e^{i\omega t} = 0,$$

$$B(i\omega)^2 e^{i\omega t} + \omega_0^2 B e^{i\omega t} - \Delta\omega^2 A e^{i\omega t} = 0,$$

or

$$(-\omega^2 + \omega_0^2)A - \Delta\omega^2 B = 0, \quad (11.31)$$

$$-\Delta\omega^2 A + (-\omega^2 + \omega_0^2)B = 0.$$

These two equations are a system of homogeneous coupled linear *algebraic* equations in the unknowns A and B . (Solving them will indirectly lead to a solution for ω .)

The technique for solving systems of coupled algebraic equations is to apply Cramer's rule. I believe you are familiar with this rule, but I will review it briefly anyway.

Consider the pair of linear algebraic equations

$$ax + by = e,$$

$$cx + dy = f.$$

